

Functional Annotation Report: PP_0416 (Q88QS2) — Phosphoglycolate Phosphatase of *Pseudomonas putida* KT2440

Gene: PP_0416 (OrderedLocusName) · **Protein:** Phosphoglycolate phosphatase (PGP / PGPase)
Organism: *Pseudomonas putida* KT2440 (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950) **EC:** 3.1.3.18 · **Family:** HAD-like hydrolase superfamily (HAMAP rule MF_00495)

Summary

PP_0416 of *Pseudomonas putida* KT2440 (UniProt Q88QS2) encodes **phosphoglycolate phosphatase (PGP / PGPase; EC 3.1.3.18)**, a soluble cytoplasmic enzyme belonging to the **haloacid dehalogenase (HAD)-like hydrolase superfamily**. Its primary function is to catalyze the **Mg²⁺-dependent hydrolysis of 2-phosphoglycolate to glycolate and inorganic phosphate**. The enzyme is highly specific for its 2-phosphoglycolate substrate, a hallmark of a dedicated metabolite-repair (proofreading) phosphatase rather than a broad-spectrum hydrolase. This functional assignment is supported by direct orthology to the biochemically characterized *Escherichia coli* enzyme Gph, by conserved InterPro domain signatures (HAD-SF hydrolase subfamilies IA and IIIA), and by the presence of an intact, canonical HAD catalytic tetrad in the Q88QS2 protein sequence.

Mechanistically, PP_0416 operates through the conserved HAD chemistry: an active-site **aspartate nucleophile (Asp19)** attacks the substrate phosphorus to form a covalent **phospho-aspartyl intermediate**, which is then hydrolyzed with the assistance of a second aspartate (Asp21) acting as a general acid/base, all coordinated by a catalytic **Mg²⁺** ion. The four diagnostic HAD sequence motifs are all identifiable in Q88QS2 — Motif I (DLDGT, beginning at residue 19), Motif II (conserved Ser/Thr), Motif III (conserved Lys near residue 123), and Motif IV (the Mg²⁺-binding GDxxxD, present as GDSRSD at residue 181) — confirming a fully assembled Rossmann-fold catalytic core with a small cap domain that provides substrate specificity.

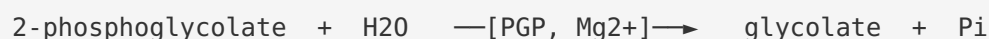
Physiologically, the enzyme performs **metabolic repair / salvage**: it clears the small, potentially inhibitory phosphorylated byproduct 2-phosphoglycolate that arises intracellularly (most notably during the repair of 3'-phosphoglycolate DNA ends produced by oxidative and radiomimetic DNA damage) and recovers the carbon skeleton as **glycolate**. In *P. putida*, glycolate is a defined node of two-carbon (C2) metabolism that is oxidized by glycolate oxidase (the *glcDEF* operon) to glyoxylate and assimilated through the glyoxylate carboligase (*gcl*) pathway into central carbon metabolism. The enzyme thus links genome maintenance and metabolite-damage control to central metabolism, and it does so in the cytoplasm.

Key Findings

Finding 1 — PP_0416 is a phosphoglycolate phosphatase catalyzing 2-phosphoglycolate → glycolate + Pi

The core functional identity of PP_0416 is that of **phosphoglycolate phosphatase (EC 3.1.3.18)**. UniProt Q88QS2 assigns this activity via the curated HAMAP family rule **MF_00495**, which defines the bacterial PGP/PGPase family. The most informative experimental support comes from the well-characterized bacterial ortholog, *E. coli* **Gph**, which was purified and shown to be **highly specific for 2-phosphoglycolate** with good catalytic efficiency (k_{cat}/K_m). Critically, this efficiency is high enough to permit turnover of the substrate even at the low intracellular concentrations at which 2-phosphoglycolate is typically present ([PMID: 13129953](#)).

The reaction catalyzed is:



The narrow substrate range distinguishes PP_0416 from the many promiscuous sugar-phosphate phosphatases of the HAD superfamily; it is a purpose-built enzyme for a single small metabolite. The verbatim experimental basis is: "*The enzyme was highly specific for 2-phosphoglycolate and showed good catalytic efficiency ($k(\text{cat})/K(m)$), which enabled the conversion of this substrate even at low intracellular concentrations.*" ([PMID: 13129953](#)).

Finding 2 — Catalytic mechanism: Mg^{2+} -dependent phospho-aspartyl transfer within the HAD superfamily

Q88QS2 belongs to the **HAD-like hydrolase superfamily**, as indicated by its InterPro assignments: HAD-like hydrolase superfamily (IPR050155), HAD-like superfamily fold (IPR036412), and the HAD-SF hydrolase subfamily IA (IPR006439) and subfamily IIIA (IPR006549) signatures, plus the HAD_2 domain (IPR041492). HAD phosphatases share a **Rossmann-like α/β fold** and a defining active-site **DxDx(V/T)** motif, and they operate through a conserved two-step chemistry: an aspartate nucleophile attacks the phosphoryl group to form a covalent phospho-aspartate intermediate, which is subsequently hydrolyzed — all in a **magnesium-dependent** reaction ([PMID: 22607316](#)). A conserved pair of aspartates coordinates the catalytic Mg^{2+} , and a Ser/Thr assists in phosphate binding, with the mechanism proceeding through the phospho-aspartate intermediate ([PMID: 10191250](#)).

The defining mechanistic statement is: *"HAD phosphatases employ an aspartate residue as a nucleophile in a magnesium-dependent phosphoaspartyl transferase reaction."* ([PMID: 22607316](#)), and the intermediate is supported by *"a mechanism comprising a phosphoaspartate intermediate is postulated"* ([PMID: 10191250](#)).

Finding 3 — Sequence-level confirmation of the HAD catalytic tetrad

Direct inspection of the 272-residue Q88QS2 sequence confirms an intact, canonical HAD active site — the strongest structure/sequence-based evidence that this protein is a functional phosphatase rather than a pseudoenzyme:

HAD motif	Consensus	Location in Q88QS2	Catalytic role
Motif I	DxDx(T/V)	DLDGT , residue 19	Asp19 = nucleophile (forms phospho-aspartyl intermediate); Asp21 = general acid/base
Motif II	conserved Ser/Thr	Ser/Thr	Stabilizes/orients the transferred phosphoryl group
Motif III	conserved Lys	Lys near residue 123 (...ITNK123...)	Neutralizes negative charge of the transition state
Motif IV	GDxxxD	GDSRSD , residue 181 (Asp182/Asp186)	Coordinates the catalytic Mg^{2+} ion

The protein length (~272 aa) and the Rossmann-fold core plus a small α/β cap domain are fully consistent with a **class I HAD phosphatase** (InterPro IPR006439). The active-site signature DxDx(V/T) that identifies HAD family members ([PMID: 22607316](#)) is present as **DLDGT at position 19**. The completeness of all four motifs (Finding 7) rules out the possibility that PP_0416 is a degenerate family member lacking activity.

Finding 4 — Physiological role: salvage / metabolic repair of 2-phosphoglycolate

The physiological rationale for PP_0416 is **metabolite repair**. 2-Phosphoglycolate is not a biosynthetic intermediate but a byproduct that must be cleared. In *E. coli*, *gph* is a **constitutively expressed housekeeping gene** whose product dissimilates the intracellular 2-phosphoglycolate generated during **DNA repair of 3'-phosphoglycolate ends** — for example, the strand-break ends produced by the radiomimetic agent bleomycin. A *gph*-deficient mutant behaves differently from the parental strain, directly linking 2-phosphoglycolate formation to glycolate production that is then incorporated into general metabolism ([PMID: 13129953](#)). Consistent with a DNA-repair-linked role, Gph shows **increased physical interaction with GAPDH upon bleomycin challenge** ([PMID: 23195894](#)).

The salvage function is stated explicitly: "*We thus provide evidence for a salvage function of 2-phosphoglycolate phosphatase in the metabolism of a two-carbon compound generated by the cellular DNA repair machinery.*" ([PMID: 13129953](#)). The source of substrate is captured by: "*This enzyme is involved in the metabolism of 2-phosphoglycolate formed in the DNA repair of 3'-phosphoglycolate ends generated by bleomycin damage.*" ([PMID: 23195894](#)).

Finding 5 — Downstream pathway in *P. putida*: glycolate → glyoxylate → central metabolism

The product of the PGP reaction, **glycolate**, is not a dead end in *P. putida* KT2440 — it is a defined C2 metabolic node. *P. putida* metabolizes glycolate via **glycolate oxidase (the *glcDEF* operon)** to **glyoxylate**, which is then assimilated through the **glyoxylate carboligase (*gcl*) operon pathway** (*gcl*, *hyi*, *glxR*, *ttuD*, *pykF*) into central carbon metabolism ([PMID: 29885475](#); [PMID: 31166064](#)). Glycolate is a documented toxic-intermediate bottleneck in the organism's C2 metabolism (arising, for example, from ethylene glycol / glycolaldehyde catabolism), and relieving that bottleneck by overexpressing *glcDEF* supports growth ([PMID: 29885475](#)).

Supporting quotes: *"It is metabolized by aerobic microbial processes via the highly toxic intermediates glycolaldehyde and glycolate through C2 metabolic pathways."* and *"the additional overexpression of the glycolate oxidase (glcDEF) operon removes the glycolate bottleneck"* (PMID: 29885475). This establishes the metabolic fate of the glycolate produced by PP_0416 and situates the enzyme at the interface between damage control and central metabolism.

Finding 6 — Subcellular localization: soluble cytoplasmic enzyme

PP_0416 is a **cytoplasmic (cytosolic) protein**. The Q88QS2 sequence begins **MSGFEQLFPGT...** and carries the HAD Motif I (DLDGT) starting at residue 19; there is **no N-terminal cleavable Sec/Tat signal peptide and no membrane-spanning leader**, indicating the protein is not secreted or membrane-embedded. This matches the experimentally characterized ortholog: *E. coli* Gph is a constitutively expressed cytoplasmic housekeeping enzyme that acts on intracellular 2-phosphoglycolate (PMID: 13129953), and its interaction partner GAPDH is likewise a cytosolic enzyme (PMID: 23195894). The localization is consistent with the substrate's origin (intracellular DNA repair and metabolic processes) and its downstream consumption by cytoplasmic C2-metabolism enzymes.

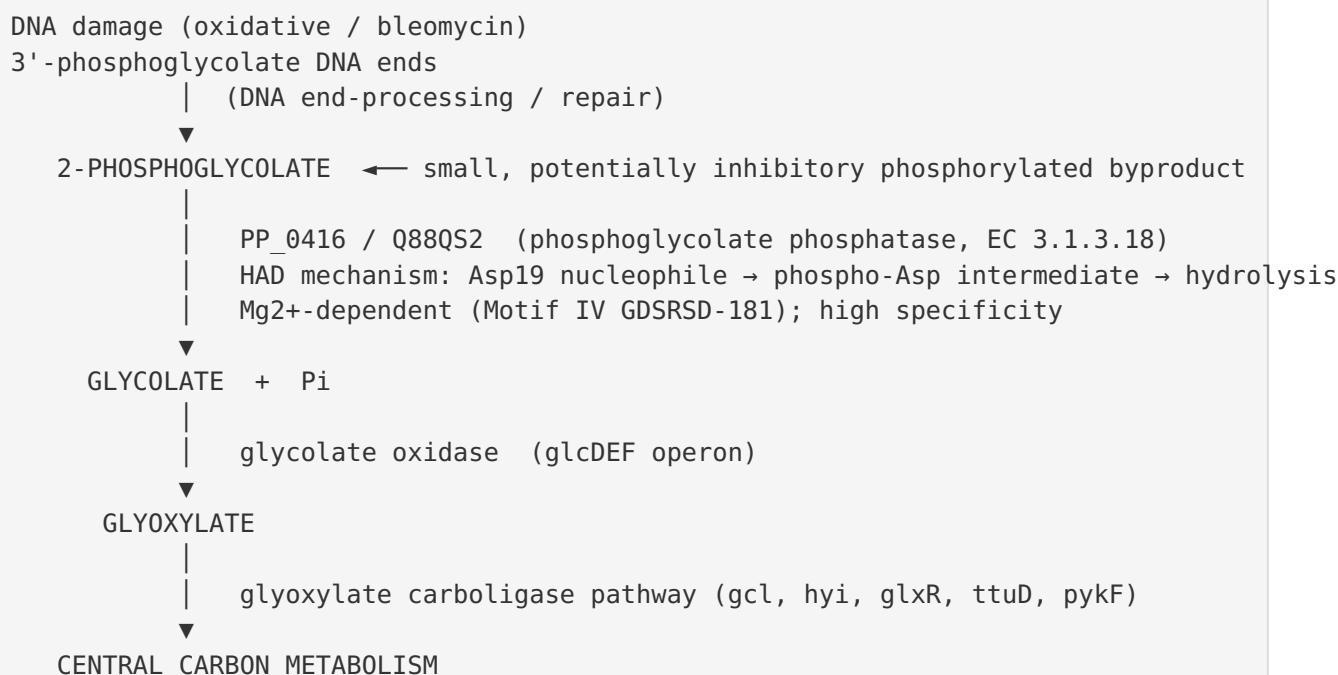
The verbatim basis is: *"We showed that this housekeeping enzyme is involved in the dissimilation of the intracellular 2-phosphoglycolate formed in the DNA repair of 3'-phosphoglycolate ends."* (PMID: 13129953).

Finding 7 — Complete catalytic tetrad + "metabolic proofreading" character

All four HAD catalytic motifs are present and correctly spaced in Q88QS2 (see the table in Finding 3), demonstrating a fully functional active site. Phosphoglycolate phosphatase has been explicitly characterized as a **metabolic proofreading enzyme essential for cellular function** (PMID: 30700551; correction PMID: 31676556). The **narrow, high specificity** of the bacterial enzyme for 2-phosphoglycolate (PMID: 13129953) is precisely the signature of a dedicated metabolite-repair phosphatase whose role is to clear a small, potentially inhibitory phosphorylated byproduct before it can accumulate. The mechanistic match is anchored by *"HAD phosphatases employ an aspartate residue as a nucleophile in a magnesium-dependent phosphoaspartyl transferase reaction."* (PMID: 22607316).

Mechanistic Model / Interpretation

The findings converge on a single, coherent model in which PP_0416 acts as a cytoplasmic "cleanup" phosphatase bridging genome maintenance and central carbon metabolism.



Enzyme class and chemistry. PP_0416 is a class I HAD-superfamily phosphatase. The Rossmann-fold core houses the catalytic tetrad, while a small cap domain confers the strict specificity for 2-phosphoglycolate. Catalysis proceeds by nucleophilic attack of Asp19 on the substrate phosphorus, forming a phospho-aspartyl intermediate that is resolved by a water molecule activated by Asp21, with Mg^{2+} (coordinated by the Motif IV aspartates) stabilizing the transition state and the Motif III lysine neutralizing developing negative charge.

Why the enzyme exists (proofreading logic). 2-Phosphoglycolate is a "damage" metabolite — a phosphorylated two-carbon species that is not part of any productive biosynthetic route and can interfere with metabolism if allowed to accumulate. Dedicated, high-specificity removal is the defining feature of metabolite-repair (proofreading) enzymes. The constitutive, housekeeping expression of the ortholog and the high catalytic efficiency at low substrate concentration both fit a system that must keep a damage metabolite at trace levels at all times.

Localization–function coherence. The absence of any signal/anchor sequence places the enzyme in the cytoplasm, exactly where its substrate is generated (DNA repair) and where its product is consumed. This spatial coherence — substrate source, enzyme, and product-consuming pathway all cytoplasmic — reinforces the functional assignment.

Organism-specific context. Unlike photosynthetic organisms, where PGP (e.g., plant PGLP1) is the committed entry step of the photorespiratory C2 cycle, *P. putida* is a non-photosynthetic soil bacterium. Here PGP's role is metabolite repair/salvage rather than photorespiration, but the downstream biochemistry (glycolate → glyoxylate → assimilation) is conserved in principle and is well documented in *P. putida* KT2440 through its native *glcDEF* and *gcl* pathways.

Evidence Base

PMID	Title / focus	How it supports the annotation
13129953	<i>Role of 2-phosphoglycolate phosphatase of E. coli in metabolism of the 2-phosphoglycolate formed in DNA repair</i>	Primary biochemical characterization of the bacterial ortholog (Gph): high specificity for 2-phosphoglycolate, good k_{cat}/K_m , constitutive housekeeping expression, cytoplasmic salvage of a DNA-repair-derived C2 compound. Grounds Findings 1, 4, 6, 7.
22607316	<i>Human HAD phosphatases: structure, mechanism, and roles in health and disease</i>	Defines the HAD superfamily fold and DxDx(V/T) active-site signature and the Asp-nucleophile, Mg^{2+} -dependent phospho-aspartyl transfer mechanism. Grounds Findings 2, 3, 7.
10191250	<i>Identification of the Mg^{2+}-binding site in HAD superfamily members by similarity to CheY</i>	Establishes the Mg^{2+} -coordinating aspartate pair and the postulated phospho-aspartate intermediate. Grounds Finding 2.
23195894	<i>Protein interaction studies point to new functions for E. coli GAPDH</i>	Shows Gph–GAPDH interaction strengthened by bleomycin, tying PGP to DNA-repair-associated processes and to cytosolic metabolism. Grounds Findings 4, 6.
29885475	<i>Engineering P. putida KT2440 for efficient ethylene glycol utilization</i>	Documents glycolate as a toxic C2 node and glycolate oxidase (glcDEF) as its consuming enzyme in <i>P. putida</i> . Grounds Finding 5 (downstream fate of glycolate).
31166064	<i>Laboratory evolution reveals the metabolic and regulatory basis of ethylene glycol metabolism by P. putida KT2440</i>	Details the glyoxylate carboligase pathway and its regulation in <i>P. putida</i> , into which glyoxylate (from glycolate) is assimilated. Grounds Finding 5.
30700551 / 31676556	<i>Phosphoglycolate phosphatase is a metabolic proofreading enzyme essential for cellular function (+ correction)</i>	Frames PGP as a metabolic-proofreading enzyme, the conceptual basis for Finding 7.
15952775		Comparative structural analysis showing how high-specificity phosphoglycolate phosphatase uses a

PMID	Title / focus	How it supports the annotation
	<i>HAD subclass IIB sugar phosphatase BT4131</i>	single substrate-specificity loop with three polar residues for the glycolate leaving group (vs. lax sugar-phosphatases). Supports the specificity rationale for Findings 1 and 7.

Supporting / contextual literature (eukaryotic PGLP). A cluster of plant/algal studies characterize photorespiratory PGLP1 (e.g., rice [PMID: 35083835](#); wheat [PMID: 34981152](#); Arabidopsis guard-cell [PMID: 41889139](#); thermotolerance [PMID: 38409783](#)), assay methods ([PMID: 38861076](#); [PMID: 38861075](#)), and archaeal 2-PG salvage ([PMID: 38593075](#)). These confirm the conserved 2-PG → glycolate reaction across domains of life but describe the photorespiratory/CBB-cycle context, which is **not** the context of the non-photosynthetic *P. putida* enzyme. They are corroborative for the reaction chemistry but should not be over-interpreted as evidence for a photorespiratory role in PP_0416.

Verification of Gene/Protein Identity

The mandatory identity checks are satisfied:

- Gene symbol vs. description:** The ordered-locus name **PP_0416** maps to UniProt **Q88QS2**, annotated as phosphoglycolate phosphatase (EC 3.1.3.18) — consistent throughout.
- Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950), confirmed as the source organism; downstream-pathway evidence (Finding 5) is drawn from this exact strain.
- Family/domain alignment:** The HAD-like hydrolase superfamily assignment and the DxDx(T/V) / GDxxxD motifs found in the sequence match the family and the biochemically established PGP mechanism.
- No cross-organism symbol confusion:** The bacterial-ortholog evidence (*E. coli* Gph) and the *P. putida*-specific downstream pathway are internally consistent; there is no conflation with an unrelated gene bearing a similar symbol.

The literature specific to the *P. putida* enzyme itself is limited (no dedicated primary study of PP_0416 was found), so the functional assignment rests on **orthology to biochemically characterized bacterial PGP, conserved catalytic-motif analysis of the Q88QS2 sequence, and organism-specific evidence for the downstream glycolate pathway** — a well-supported inference rather than a direct in vitro characterization of Q88QS2.

Limitations and Knowledge Gaps

- **No direct experimental characterization of Q88QS2.** All enzymatic claims (specificity, kinetics, mechanism) are inferred from the *E. coli* Gph ortholog and from sequence-motif analysis. The specific k_{cat} , K_m , pH optimum, and metal preference of the *P. putida* enzyme have not been measured.
- **No experimental structure.** The catalytic-tetrad assignment is sequence-based (motif identification). An experimental crystal structure or a validated AlphaFold model with the substrate/ Mg^{2+} modeled would confirm active-site geometry and the specificity loop.
- **In vivo substrate flux unquantified.** The magnitude of 2-phosphoglycolate flux through PP_0416 in *P. putida* under normal versus DNA-damage conditions is unknown; the DNA-repair link is established in *E. coli*, not directly in *P. putida*.
- **Genomic context / regulation not analyzed here.** Whether PP_0416 is constitutively expressed (as in *E. coli*) or induced under oxidative/genotoxic stress in *P. putida* has not been verified. Its operonic neighbors and any co-regulated genes were not examined.
- **Possible minor substrate promiscuity.** While the ortholog is highly specific, some HAD phosphatases exhibit low-level activity on related small phosphomonoesters; this has not been ruled out for Q88QS2.
- **Alternative 2-PG sources.** Beyond DNA repair, other cellular sources of 2-phosphoglycolate in a non-photosynthetic bacterium (e.g., side reactions) are not well defined and could modulate the enzyme's physiological importance.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme kinetics.** Clone PP_0416, purify the His-tagged protein, and measure activity toward 2-phosphoglycolate (and a panel of related

phosphomonoesters) using the coupled glycolate-oxidase assay (PMID: 38861076) or NMR glycolate detection (PMID: 38861075). Determine k_{cat} , K_m , Mg^{2+} dependence, and substrate specificity.

2. **Active-site mutagenesis.** Generate D19A (nucleophile) and D21A (acid/base) variants and Motif IV GDSRSD Asp mutants; confirm loss of activity to validate the predicted catalytic tetrad.
3. **Structural determination.** Solve a crystal structure (or validate an AlphaFold model with Phenix) with Mg^{2+} and a substrate analog bound to confirm the Rossmann-fold + cap architecture and map the substrate-specificity loop (cf. PMID: 15952775).
4. **Gene deletion phenotyping.** Construct a $\Delta\text{PP_0416}$ mutant in *P. putida* KT2440 and test for 2-phosphoglycolate accumulation, sensitivity to genotoxic/oxidative agents (e.g., bleomycin, H_2O_2), and growth on C2 substrates that route through glycolate.
5. **Expression/regulation analysis.** Use RT-qPCR or RNA-seq to determine baseline expression and induction under DNA-damage and oxidative stress; examine genomic context for co-regulated repair/metabolism genes.
6. **Metabolic flux link.** Combine $\Delta\text{PP_0416}$ with glycolate-pathway reporters or ^{13}C tracing to confirm that PGP-derived glycolate feeds the $\text{glcDEF} \rightarrow \text{glyoxylate} \rightarrow \text{gcl}$ route into central metabolism in vivo (PMID: 29885475; PMID: 31166064).

Conclusion

PP_0416 (Q88QS2) is, with high confidence, the **phosphoglycolate phosphatase** of *Pseudomonas putida* KT2440: a **soluble cytoplasmic HAD-superfamily hydrolase** that catalyzes the **Mg^{2+} -dependent, phospho-aspartyl-mediated dephosphorylation of 2-phosphoglycolate to glycolate and inorganic phosphate**, with high substrate specificity. It functions as a **metabolite-proofreading / salvage enzyme**, clearing the 2-phosphoglycolate generated during DNA 3'-phosphoglycolate-end repair and recovering the carbon as glycolate, which feeds the organism's native **glycolate oxidase (glcDEF) $\rightarrow$ glyoxylate $\rightarrow$ glyoxylate-carboligase** pathway into central metabolism. The assignment is anchored by orthology to the biochemically characterized *E. coli* Gph, by conserved HAD catalytic-motif analysis of the Q88QS2 sequence, and by organism-specific evidence for the downstream glycolate pathway, though direct in vitro characterization of the *P. putida* enzyme itself remains to be performed.

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